Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years designing, developing, testing, and releasing product features using Java, JavaScript, TypeScript, Angular, React, SQL, and REST APIs. Experienced in Agile delivery, relational databases, debugging, automated testing, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built REST APIs and Azure SQL services for 500+ users, translating product requirements into tested features while optimizing schemas and queries to reduce response latency from 1.2 seconds to 350 milliseconds.
- Refactored a legacy workflow into maintainable services with explicit states, automated validation, and failure isolation, tripling throughput while improving code review, debugging, release reliability, and system documentation.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed education-focused Flask REST services handling 10,000+ daily requests, implementing authentication, grading workflows, validation, and error handling while debugging production issues affecting more than 150 users.
- Delivered responsive React workflows and multithreaded processing for 1,000+ files with relational persistence, automated tests, progress monitoring, and recovery, reducing feedback turnaround 60% under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python pipelines processing 100,000+ records, improving planning accuracy 18% through reusable components, automated validation, structured experiments, performance monitoring, detailed technical documentation, and systematic debugging.
- Collaborated with researchers to translate ambiguous requirements into modular software, improving closed-loop stability 15% while documenting designs, reviewing results, and validating expected behavior across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed modular services and SQL integrations using AWS Glue, S3, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68% across connected business applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating technical issue investigation, regression verification, peer code reviews, and reliable releases across multiple environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed JavaScript and Node.js services across eight microservices with authentication, REST APIs, WebSocket messaging, and state synchronization, securely processing more than two million Azure SQL database records.
- Automated builds, functional tests, staged deployments, and monitoring through Jenkins, Docker, and Azure, reducing release time 50% while improving validation coverage, release consistency, and overall operational visibility.

Skills

Languages & Frontend: Java, JavaScript, TypeScript, Python, SQL, React, Angular.js, Next.js, HTML, CSS, object-oriented programming, data structures, algorithms
Backend & Data: Node.js, Flask, FastAPI, REST APIs, WebSockets, microservices, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, relational database design, query optimization
Testing & Quality: Unit testing, integration testing, automated validation, PyTest, Mockito, debugging, log analysis, performance monitoring, regression testing, authentication, role-based access
SDLC & Delivery: Git, code reviews, Jenkins, GitHub Actions, CI/CD, Docker, Azure, Jira, Agile, planning, estimation, technical documentation, release processes, cross-functional collaboration

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a modular Java and TypeScript database tool supporting 10+ engines with schema-aware SQL generation, query validation, metadata exploration, error handling, automated testing, and Docker-backed CI/CD deployment.

PiXMod | Live | *Angular.js, Next.js, JavaScript, MongoDB, Stripe*
- Built a responsive full-stack product using Angular.js, Next.js, MongoDB, and Stripe, implementing maintainable user workflows, API integrations, automated validation, and subscription features that increased premium conversion 18%.

HMIS | Live | *MERN, TypeScript, REST APIs, Sentry, Agile*
- Built a TypeScript workflow platform with REST APIs, role-based access, scheduling, notifications, and Sentry monitoring, reducing administrative workload 30% through tested features, responsive dashboards, and documented releases.